

ESAT CAMP MOCK E

Full sitting

EC-FULL-E

MATHEMATICS 1

40 MINUTES

MATHEMATICS 2

40 MINUTES

PHYSICS

40 MINUTES

CHEMISTRY

40 MINUTES

BIOLOGY

40 MINUTES

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin**.

This paper contains **5 modules** (Mathematics 1, Mathematics 2, Physics, Chemistry, Biology). Each module has **27 multiple-choice questions** and is designed for **40 minutes** (about **200 minutes** in total if you sit every module).

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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Mathematics 1

- 1 A curve has the equation $y^2 = kx$ for some constant $k > 0$. The point $P(a, b)$, where $a > 0$ and $b > 0$, lies on the curve. The point Q , which also lies on the curve, has an x -coordinate of $9a$ and a positive y -coordinate. What is the y -coordinate of Q in terms of b ?

- A $3b$
- B $9b$
- C $81b$
- D $\frac{b}{3}$
- E $\frac{b}{9}$
- F $3b^2$

- 2 Which one of the following is a simplification of

$$\frac{2x}{x-3} - \frac{x^2 + 3x}{9 - x^2} - 1$$

for all values of x for which the expression is defined?

- A $2 + \frac{9}{x-3}$
- B $2 - \frac{9}{x-3}$
- C $2 + \frac{3}{x-3}$
- D $2 - \frac{3}{x-3}$
- E $1 + \frac{3}{x-3}$
- F $1 - \frac{3}{x-3}$

- 3 Find the value of x that satisfies the equation:

$$4^{\frac{1}{x-3}} \times 2^{\frac{1}{x+3}} = 16^{\frac{x-2}{x^2-9}}$$

- A $-\frac{5}{2}$
- B $\frac{5}{7}$
- C $\frac{5}{2}$
- D 4
- E 5
- F 11

- 4** A material is composed of a mixture of two types of spherical particles, type A and type B. The ratio of the number of particles of type A to the number of particles of type B is 4 : 1. The total volume occupied by all particles of type A is half the total volume occupied by all particles of type B. What is the ratio of the radius of a single particle of type A to the radius of a single particle of type B?
- A** $\frac{1}{8}$
B $\frac{\sqrt{2}}{4}$
C $\frac{1}{2}$
D $\sqrt{2}$
E 2
F 8
- 5** A point P lies on the hypotenuse AB of a right-angled triangle ABC . The perpendicular distance from P to the leg AC is equal to the perpendicular distance from P to the leg BC . Given that $AP = 3$ and $PB = 4$, what is this distance?
- A** $12/7$
B $12/5$
C 2.5
D $2\sqrt{3}$
E $7\sqrt{2}/4$
- 6** A solid cuboid has side lengths 2^a , 2^b , and 2^c , where a, b, c are integers such that $a > b > c > 0$. A process is applied repeatedly: the cuboid is cut in half by a plane perpendicular to its longest edge. If there is more than one longest edge, only one is chosen to be cut. The process stops as soon as the cuboid is a cube. Let N be the total number of cuts required, and let S be the side length of the final cube. Which pair of expressions for N and S is correct?
- A** $N = a - b, \quad S = 2^b$
B $N = a - c, \quad S = 2^c$
C $N = a + b - c, \quad S = 2^c$
D $N = a + b - 2c, \quad S = 2^c$
E $N = a + b - 2c, \quad S = 2^{3c}$

- 7 A tank initially contains V litres of water. Water flows into the tank at a rate of x litres per minute for a duration of h hours. Water flows out of the tank at a rate of y litres per hour for a duration of m minutes.
Which of the following expressions gives the final volume of water in the tank, in litres?
- A $V + hx - my$
 B $V + 60hx - 60my$
 C $V + \frac{hx}{60} - \frac{my}{60}$
 D $V + 60hx - \frac{my}{60}$
 E $V + \frac{hx}{60} - 60my$
 F $V - 60hx + \frac{my}{60}$
- 8 The points $A(2, 6)$ and $B(6, -2)$ lie on a straight line. Point P lies on this line such that the distance AP is three times the distance PB . Given that P does not lie between A and B , what are the coordinates of P ?
- A $(0, 10)$
 B $(3, 4)$
 C $(5, 0)$
 D $(8, -6)$
 E $(14, -18)$
- 9 A passcode of length 5 is formed by selecting distinct digits from the set $\{1, 2, 3, 4, 5, 6\}$.
The digit 6 must be included in the passcode, but it cannot be the first or the last digit.
How many such passcodes can be formed?
- A 120
 B 240
 C 360
 D 480
 E 600
 F 720

- 10** A particle undergoes two successive displacements. The first displacement has magnitude d and is on a bearing of 030° . The second displacement has magnitude $6\sqrt{3}$ and is on a bearing of 300° .
Given that the final position of the particle is due north of its initial position, what is the value of d ?
- A** 6
B 9
C $6\sqrt{3}$
D 18
E $12\sqrt{3}$
- 11** A composite object is formed from two materials, X and Y. The density of material X is 2000 kg m^{-3} . The density of material Y is 8.0 g cm^{-3} . The volume of material X is twice the volume of material Y.
What is the density of the composite object?
- A** 1600 kg m^{-3}
B 4000 kg m^{-3}
C 5000 kg m^{-3}
D 6000 kg m^{-3}
- 12** A solid cylinder has base radius r and height h . A solid cube has side length $2r$.
The total surface area of the cylinder is equal to the total surface area of the cube.
Which of the following expressions gives h in terms of r ?
- A** $h = \left(\frac{8}{\pi} - 1\right)r$
B $h = \frac{8r}{\pi}$
C $h = \left(\frac{12}{\pi} - 1\right)r$
D $h = \frac{12r}{\pi}$
E $h = \left(\frac{12}{\pi} + 1\right)r$

- 13** Three fair, six-sided dice are rolled. Let p_S be the probability that the sum of the scores is 5. Let p_P be the probability that the product of the scores is 6. What is the value of $p_P - p_S$?

- A** 0
- B** $\frac{1}{216}$
- C** $\frac{1}{108}$
- D** $\frac{1}{72}$
- E** $\frac{1}{36}$

- 14** Two regular polygons, P and Q, are such that the exterior angle of P is three times the exterior angle of Q.

The sum of an interior angle of P and an interior angle of Q is 300° .

What is the number of sides of polygon Q?

- A** 6
- B** 8
- C** 12
- D** 24
- E** 30
- F** 48

- 15** A large square with side length $\sqrt{21} + \sqrt{3}$ contains four identical shaded rectangles arranged around a central unshaded square of side length $\sqrt{21} - \sqrt{3}$, as shown in the diagram.
- What is the total area of the shaded region?

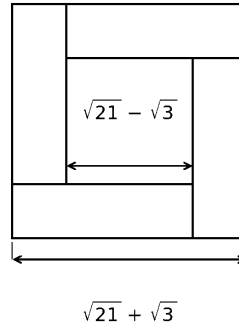


Diagram not to scale

- A** $3\sqrt{7}$
- B** $6\sqrt{7}$
- C** $12\sqrt{3}$
- D** $12\sqrt{7}$
- E** 24
- F** $24\sqrt{7}$
- G** 48

- 16** The diagram shows a circle of radius r inscribed inside a sector of a circle of radius R . The central angle of the sector is 2θ , where $0 < \theta < \frac{\pi}{2}$. The inscribed circle is tangent to both straight edges of the sector and to the circular arc. Which of the following is an expression for r in terms of R and θ ?

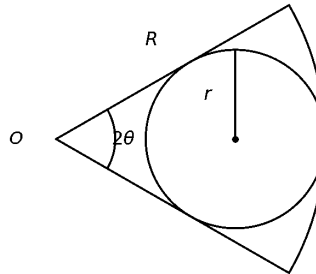


Diagram not to scale

- A** $\frac{R \sin \theta}{1 + \sin \theta}$
B $\frac{R \sin \theta}{1 - \sin \theta}$
C $\frac{R \cos \theta}{1 + \cos \theta}$
D $\frac{R \cos \theta}{1 - \cos \theta}$
E $\frac{R \tan \theta}{1 + \tan \theta}$
F $\frac{R}{1 + \sin \theta}$
- 17** The pressure of a gas sample is increased by 25%. The final pressure is 120 kPa. What was the initial pressure?
- A** 90 kPa
B 95 kPa
C 96 kPa
D 100 kPa
E 150 kPa

- 18** A list consists of 6 distinct numbers. A new number is added to the list, creating a new list with 7 numbers.

The mean of the new list is identical to the mean of the original list.

Which one of the following statements must be true?

- A** The range is unchanged.
 - B** The median is unchanged.
 - C** The range increases.
 - D** The median is now equal to the mean.
 - E** The median increases.
 - F** The median decreases.
- 19** Given that p and q are real numbers such that

$$p + q = 1$$

and

$$8^p \times 27^q = 12$$

what is the value of p ?

- A** $\frac{1}{3}$
- B** $\frac{4}{19}$
- C** $\frac{1}{2}$
- D** $\frac{15}{19}$
- E** $\frac{2}{3}$

20 Sets of non-zero integers are classified as Type A or Type B.

- In a Type A set, 100% of the integers are even.
- In a Type B set, 50% of the integers are odd.

In any set of either type, 50% of the integers are positive. An integer's sign is independent of its parity.

Two disjoint sets, S_1 and S_2 , have equal size. An integer is chosen at random from the union $S_1 \cup S_2$.

What is the probability that the integer is odd and positive in Case 1, and what is this probability in Case 2?

- Case 1: Both S_1 and S_2 are Type B.
- Case 2: One of S_1, S_2 is Type A and the other is Type B.

- A** Case 1: $\frac{1}{4}$, Case 2: $\frac{1}{8}$
B Case 1: $\frac{1}{8}$, Case 2: $\frac{1}{4}$
C Case 1: $\frac{1}{2}$, Case 2: $\frac{1}{4}$
D Case 1: $\frac{1}{4}$, Case 2: $\frac{3}{8}$
E Case 1: $\frac{1}{8}$, Case 2: 0
F Case 1: $\frac{1}{4}$, Case 2: $\frac{1}{4}$

21 Element Q has three isotopes. The lightest isotope has a relative mass of 20. The isotope with the middle mass is 20% heavier than the lightest, and the heaviest isotope is 30% heavier than the lightest.

The relative abundance of the middle-mass isotope is equal to the sum of the relative abundances of the other two. The relative abundance of the lightest isotope is twice that of the heaviest isotope.

What is the relative atomic mass of Q?

- A** 22.0
B 22.5
C 23.0
D 23.5
E 24.0

- 22** Three distinct vertices are chosen at random from the eight vertices of a regular octagon. What is the probability that the triangle formed by these three vertices is right-angled?

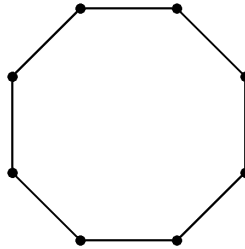


Diagram not to scale

- A** $\frac{1}{14}$
B $\frac{1}{7}$
C $\frac{2}{7}$
D $\frac{3}{7}$
E $\frac{4}{7}$
F $\frac{6}{7}$
- 23** A fluid containing a dissolved substance flows through a cylindrical pipe. Between point A and point B, the pipe's radius is halved, the fluid's speed triples, and the substance's concentration decreases by 20%. What is the percentage change in the mass of the substance flowing past a point per second, from A to B?
- A** 130% increase
B 50% increase
C 20% increase
D 25% decrease
E 40% decrease
F 60% decrease

- 24** An empty rectangular tank is being filled with water from a cylindrical pipe. The tank's internal dimensions are $5x$ m, $2x$ m, and y m. The pipe's internal diameter is $2x$ cm. The water flows at a constant speed of $\frac{1}{2}y$ km/h.

For positive constants x and y , find the time in hours required to fill the tank.

- A** $\frac{2}{\pi}$
- B** $\frac{20}{\pi}$
- C** $\frac{50}{\pi}$
- D** $\frac{200}{\pi}$
- E** $\frac{200\,000}{\pi}$
- F** $\frac{1}{50\pi}$

- 25** A circular sector has a fixed radius r and a variable angle θ radians. Its perimeter is P and its area is A .

A graph of A against P is a straight line which passes through the points $(14, 6)$ and $(20, 24)$.

What is the value of r ?

- A** $\frac{2}{3}$
- B** 3
- C** 4
- D** 6
- E** 8
- F** 12

- 26** Which inequality defines the complete set of values of x satisfying $|3 - 2x| < 9$?

- A** $-3 < x < 6$
- B** $-6 < x < 3$
- C** $x < 6$
- D** $x > -3$
- E** $x < -3$ or $x > 6$

27 Which one of the following is a simplification of $(3\sqrt{2} - 2\sqrt{3})^2$?

A 6

B 30

C $6 - 12\sqrt{6}$

D $12 - 12\sqrt{6}$

E $30 - 6\sqrt{6}$

F $30 - 12\sqrt{6}$

G $30 + 12\sqrt{6}$

H $6\sqrt{2} - 4\sqrt{3}$

Mathematics 2

- 1 The value of the definite integral

$$\int_1^2 \ln \left(\frac{e^{2x}}{x^2} \right) dx$$

is

- A** 4
B $2\ln 2$
C $3 - 4\ln 2$
D $5 - 4\ln 2$
E $1 + 4\ln 2$
- 2 Find the number of solutions to the equation

$$\log_2(\sin x) + \log_2(3 \sin x) = 1 + \log_2(1 - \cos x)$$

for $0 \leq x \leq 2\pi$.

- A** 0
B 1
C 2
D 3
E 4
F 5
- 3 P, Q and R are regular polygons.

Q has twice as many sides as P.

R has three times as many sides as Q.

An interior angle of R is 25° larger than an interior angle of P.

How much larger is an interior angle of Q than an interior angle of P, in degrees?

- A** 5
B $8\frac{1}{3}$
C 10
D $12\frac{1}{2}$
E 15
F $16\frac{2}{3}$
G 20

- 4 The function $f(x) = 2x - x^2$ is defined for $0 \leq x \leq k$, where $k > 2$. Let I be the definite integral $\int_0^k f(x) dx$. Let A be the area of the region bounded by the curve $y = f(x)$ and the x -axis for $0 \leq x \leq k$. Given that $A = 3I$, what is the value of k ?

- A 2
- B 3
- C $1 + \sqrt{2}$
- D $1 + \sqrt{3}$
- E $2 + \sqrt{3}$

- 5 Two vertices of a regular hexagon are at the points $(1, 1)$ and $(3, 5)$.

What is the sum of the areas of all possible regular hexagons with these points as vertices?

- A $\frac{95\sqrt{3}}{2}$
- B $\frac{75\sqrt{3}}{2}$
- C $\frac{105\sqrt{3}}{2}$
- D $40\sqrt{3}$
- E $\frac{35\sqrt{3}}{2}$
- F $\frac{95\sqrt{3}}{12}$

- 6 The straight line XY is tangent at point A to a circle with centre O .
 Points B and C lie on the circumference of the circle such that $AB = BC$.
 Angle $YAB = 65^\circ$.
 What is the size of angle OCA ?

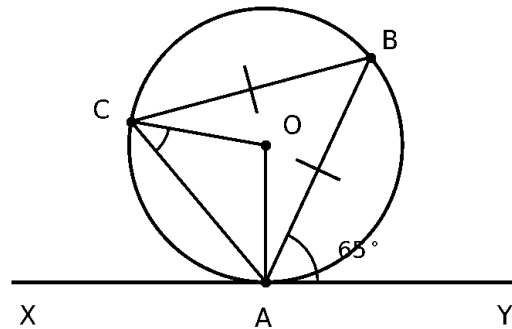


Diagram not to scale

- A 25°
 B 35°
 C 40°
 D 45°
 E 50°
 F 65°
 G 75°
 H 80°
- 7 The first term of an arithmetic sequence is 1 and its second term is $\frac{1}{6}$.
 A geometric sequence has its first term equal to the third term of the arithmetic sequence, and its common ratio equal to the sum of the first three terms of the arithmetic sequence.
 Find the sum to infinity of the geometric sequence.

- A $-\frac{4}{3}$
 B 2
 C $\frac{1}{3}$
 D $-\frac{4}{9}$
 E $-\frac{5}{3}$
 F $-\frac{3}{4}$

- 8 The constant k is defined by

$$k = \int_0^2 (3x^2 + 1) dx.$$

A geometric series has first term k and common ratio $1/3$.

What is its sum to infinity?

- A 10
- B 12
- C 15
- D 18
- E 20
- F 30

- 9 What is the maximum value of the expression $\sin^2 x + 3 \cos x$ for real values of x ?

- A 1
- B 3
- C $13/4$
- D 4
- E 5

- 10 A particle moves along a straight line such that its displacement x in metres from a fixed origin at time t in seconds is given by:

$$x = 5 \cos(3t)$$

What is the total distance traveled by the particle between $t = 0$ and $t = \frac{5\pi}{6}$?

- A 5
- B 15
- C 20
- D 25
- E 30

- 11 How many real solutions does the equation

$$\sin(2x) = \tan(x)$$

have in the interval $0 \leq x \leq 2\pi$?

- A** 3
B 4
C 5
D 6
E 7
- 12 Two independent discrete random variables, X and Y , each take values in the set $\{1, 2, 3\}$. The probability distributions of X and Y are given in the table below:

k	1	2	3
$P(X=k)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$
$P(Y=k)$	p	p	q

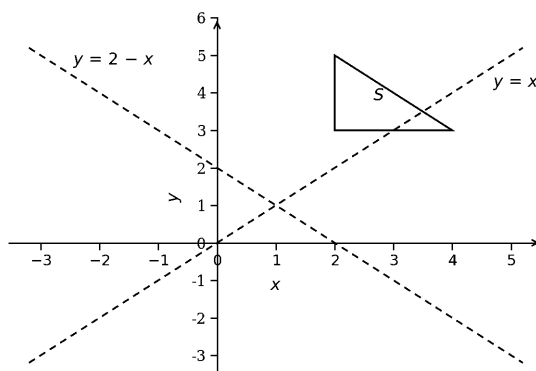
where p and q are constants.

Given that $P(X + Y = 6) = \frac{1}{12}$, what is the value of $P(X + Y = 3)$?

- A** $\frac{1}{12}$
B $\frac{1}{8}$
C $\frac{1}{6}$
D $\frac{5}{24}$
E $\frac{1}{4}$
F $\frac{5}{12}$

- 13** The shape S shown in the diagram is reflected in the line $y = x$, and the resulting image is then reflected in the line $y = 2 - x$.

Which single transformation is equivalent to this sequence of two reflections?



- A** A translation by the vector $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$
- B** A translation by the vector $\begin{pmatrix} -2 \\ -2 \end{pmatrix}$
- C** A reflection in the line $x = 1$
- D** A reflection in the line $y = 1$
- E** A rotation of 90° clockwise about the point $(1, 1)$
- F** A rotation of 90° anticlockwise about the point $(1, 1)$
- G** A rotation of 180° about the origin $(0, 0)$
- H** A rotation of 180° about the point $(1, 1)$
- 14** The area of the region enclosed by the curve $y = 3x^2 + 1$, the x -axis, and the lines $x = 0$ and $x = 2$, is A .
The area of the region enclosed by the line $y = mx$, the x -axis, and the lines $x = 0$ and $x = 2$, is 10% of A .

What is the value of m ?

- A** $\frac{1}{4}$
- B** $\frac{2}{5}$
- C** $\frac{1}{2}$
- D** 1
- E** $\frac{13}{10}$
- F** $\frac{9}{2}$

15 What is the term independent of x in the expansion of $\left(2x^2 - \frac{1}{x}\right)^6$?

- A -160
- B -60
- C 15
- D 30
- E 60
- F 240

16 The table shows summary statistics for the scores achieved by students in three different classes, X, Y, and Z.

Class	Number of students	Mean score	Range of scores
X	15	18	8
Y	25	24	14
Z	10	30	11

The results for all three classes are combined into a single dataset.

What can be deduced about the mean and range of the combined dataset?

- A mean = 23.4, range ≤ 11
- B mean = 23.4, $11 < \text{range} < 14$
- C mean = 23.4, range ≥ 14
- D mean = 23.4, range ≥ 33
- E mean = 24.0, range ≤ 11
- F mean = 24.0, $11 < \text{range} < 14$
- G mean = 24.0, range ≥ 14
- H mean = 24.0, range ≥ 33

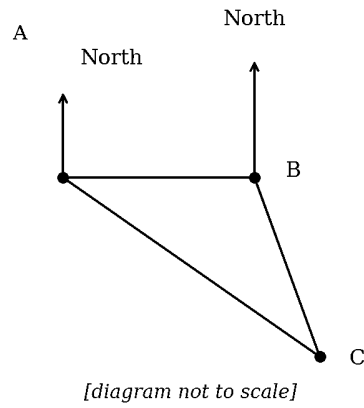
- 17 Three buoys, A , B , and C , are floating in the sea as shown in the diagram.

Buoy B is due East of buoy A .

The distance from A to B is equal to the distance from B to C .

Buoy C is on a bearing of 160° from buoy B .

What is the bearing of buoy A from buoy C ?



- A 015°
- B 125°
- C 285°
- D 305°
- E 325°
- F 340°

- 18 A dataset of 25 integer values is summarised in the frequency table below:

Value	Frequency
2	p
4	q
6	7
8	r
10	2

where p , q , and r are positive integers.

The dataset has:

- a mean of 5.44
- a median of 6
- a unique mode of 4

What is the value of p ?

- A 1
- B 2
- C 3
- D 4
- E 5
- F 7
- 19 A real number x satisfies the equation

$$\frac{e^{2x} - e^{-2x}}{e^{2x} + e^{-2x}} = \frac{1}{3}$$

What is the value of x ?

- A $\frac{1}{4} \ln 2$
- B $\frac{1}{2} \ln 2$
- C $-\frac{1}{4} \ln 2$
- D $-\frac{1}{2} \ln 2$
- E 0
- F $\ln 2$

- 20** A container is constructed in the shape of a cuboid with a square base of side length x and height h . The container is open at the top. The total surface area of the container is fixed at a constant value S . Given that the volume of the container is maximised, which of the following equations correctly relates h and x ?

- A** $h = \frac{1}{4}x$
- B** $h = \frac{1}{2}x$
- C** $h = x$
- D** $h = 2x$
- E** $h = 4x$

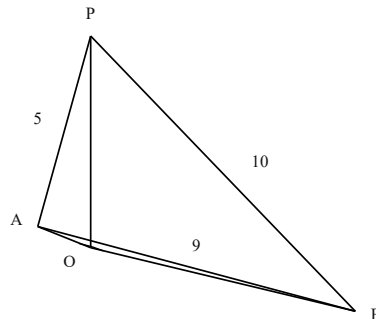
- 21** Find the coefficient of x^2 in the expansion of

$$\left(\frac{x^3 - x}{x^2 - 1} + \frac{-2x^3 - 4x^2 - 2x - 4}{x^3 + 2x^2 + x + 2} \right)^5$$

- A** -80
- B** -40
- C** -20
- D** 20
- E** 40
- F** 80

- 22** A vertical pole stands at a point O on horizontal ground. Points A and B are on the ground such that the triangle OAB is right-angled at O.

The distance from the top of the pole to A is 5, and the distance from the top of the pole to B is 10. The distance between A and B is 9.



What is the height of the pole?

- A $\sqrt{3}$
 - B $\sqrt{21}$
 - C $\sqrt{22}$
 - D $\sqrt{23}$
 - E $\sqrt{44}$
 - F $\sqrt{103}$
- 23** Three distinct numbers form the first three terms of an arithmetic progression and have a sum of 12. These three numbers, when reordered, form consecutive terms of a geometric progression. What is the value of the largest of the three numbers?
- A 4
 - B 8
 - C 12
 - D 16
 - E 20

- 24 A curve has equation

$$y = \frac{k(2x^3)^2 - 4x^5}{x}$$

for $x > 0$, where k is a constant.

The gradient of the tangent to the curve at the point where $x = 1$ is 12.

What is the value of k ?

- A $\frac{7}{5}$
- B $\frac{8}{5}$
- C $\frac{7}{4}$
- D $\frac{14}{5}$
- E 4
- F 8

- 25 What is the value of the sum

$$\sum_{n=2}^{100} \log_{10} \left(\frac{n^3 - n}{n^3 + n^2} \right)$$

- A -2
- B -1
- C 0
- D 1
- E 2

- 26 The functions $f(x)$ and $g(x)$ satisfy the equation

$$2 \int_0^3 (f(x) + 1) \, dx + \int_0^3 (4g(x) - f(x)) \, dx = 15$$

What is the value of the following integral?

$$\int_0^3 (5f(x) + 20g(x) - 2x) \, dx$$

- A 36
- B 39
- C 45
- D 51
- E 54
- F 56

27 The solution set of the inequality

$$\frac{\log_2 x - k}{\log_2 x - 1} > 3$$

is the interval $\frac{1}{4} < x < 2$.

Find the value of the constant k .

A -1

B 1

C 3

D 4

E 5

F 7

Physics

- 1 A solid block of uniform cross-sectional area A and mass m floats upright in a liquid of density ρ_0 , as shown in the diagram. The lower portion of the block is submerged to a depth h .

The liquid is then heated uniformly. The liquid expands so that its density ρ_0 decreases, while the thermal expansion of the solid block is negligible. The block remains floating upright.

Here are three statements about the system after the liquid is heated:

- 1 The submerged depth h of the block increases.
- 2 The upthrust (buoyant force) acting on the block decreases.
- 3 The liquid pressure acting on the bottom surface of the block remains unchanged.

Which of these statements is/are correct?

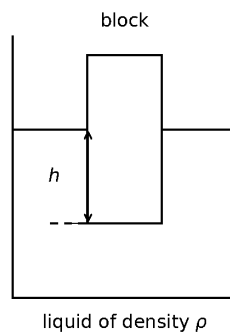


Diagram not to scale

- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 2 A radioisotope thermal generator uses a radioactive isotope P, which decays into a single stable isotope Q. The heat generated is proportional to the activity of the sample.

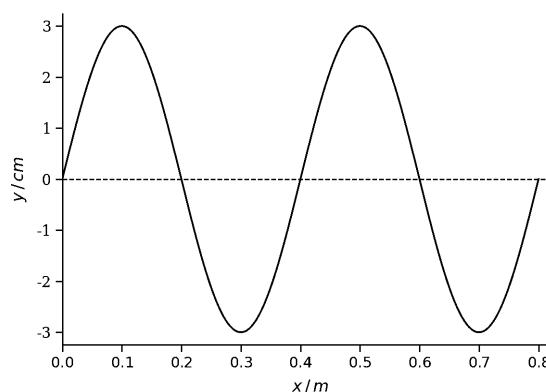
The fuel is initially pure isotope P, which has a half-life of 20 years. After 80 years of operation, what is the ratio of the number of atoms of Q to the number of atoms of P?

- A 1/16
- B 1/15
- C 7
- D 15/16
- E 15
- F 16

- 3 A continuous transverse wave travels at a constant speed of 12 m s^{-1} along a stretched string in the positive x -direction. The graph shows the displacement y of the string as a function of position x at time $t = 0$.

What is the total distance travelled by a particle of the string during a time interval of 0.50 s, and what is the distance travelled by the wave during this same time interval?

	distance travelled by particle / m	distance travelled by wave / m
A	0.45	6.0
B	0.45	12.0
C	0.90	6.0
D	0.90	12.0
E	1.8	6.0
F	1.8	12.0
G	3.6	6.0
H	3.6	12.0



- 4 An electric heater supplies energy at a constant rate of 400 W to a metal block of mass 3.0 kg. The specific heat capacity of the metal is $1000 \text{ J kg}^{-1} \text{ K}^{-1}$. The block loses heat to its surroundings at a constant rate of 100 W.

What is the time taken for the temperature of the block to increase by 20 K?

- A 120 s
- B 150 s
- C 200 s
- D 300 s
- E 600 s

- 5 Spherical catalyst particles of radius r fall through a gaseous reactant mixture. The drag force F acting on a particle moving with velocity v is given by:

$$F = kr^2v^2$$

where k is a constant.

The mass of a particle is proportional to r^3 . At the terminal velocity v_t , the drag force is equal to the gravitational force.

Which of the following gives the relationship between v_t and r ?

- A $v_t \propto r^2$
- B $v_t \propto r$
- C $v_t \propto \sqrt{r}$
- D $v_t \propto \frac{1}{\sqrt{r}}$
- E $v_t \propto r^{\frac{3}{2}}$

- 6 An astronaut of mass 100 kg is at rest in space. The astronaut throws a 4.0 kg tool, which then moves at a speed of 5.0 m/s. What is the astronaut's recoil speed?

- A 0.05 m/s
- B 0.20 m/s
- C 1.0 m/s
- D 5.0 m/s
- E 20 m/s
- F 125 m/s

- 7 A U-shaped magnet rests on an electronic balance. A rigid, horizontal straight wire of length 15 cm is clamped independently in a fixed position between the poles of the magnet, perpendicular to a uniform magnetic field of flux density 0.40 T.

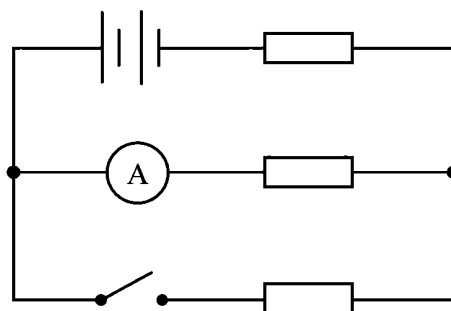
A constant current is switched on in the wire such that a total charge of 45 C passes through the wire in 2.5 minutes. This current causes a vertically upward magnetic force to act on the wire.

What is the resulting change in the reading displayed by the balance?

(Take the acceleration of free fall to be $g = 10 \text{ m s}^{-2}$.)

- A decrease of 0.18 g
- B increase of 0.18 g
- C decrease of 1.8 g
- D increase of 1.8 g
- E decrease of 18 g
- F increase of 18 g
- G decrease of 108 g
- H increase of 108 g

- 8 A circuit is set up as shown. The battery has negligible internal resistance and all three resistors are identical.



When the switch is open, the reading on the ammeter is 3.0 A and the power transferred from the battery is 18 W.

The switch is now closed.

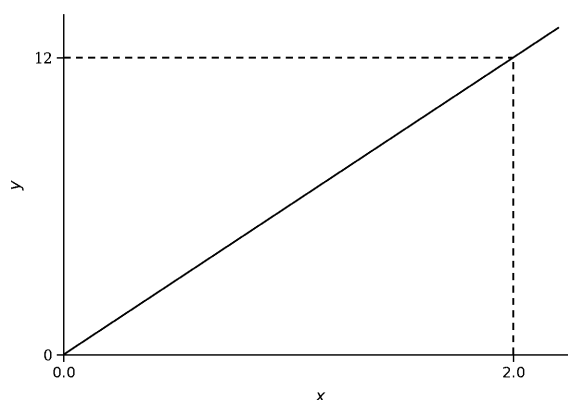
What is the new reading on the ammeter and what is the new power transferred from the battery?

	ammeter reading / A	power transferred / W
A	1.5	18
B	1.5	24
C	2.0	12
D	2.0	24
E	2.0	36
F	3.0	24
G	3.0	27
H	3.0	36

- 9** The graph shown of quantity y against quantity x represents a physical relationship, where both quantities are measured in their standard SI units.

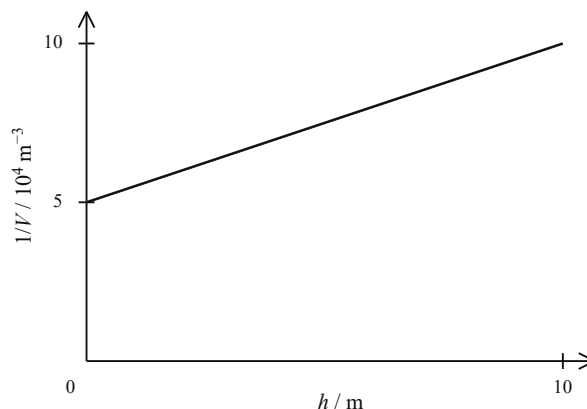
Which two of the following relationships could this graph represent?

- 1 momentum against velocity for an object of mass 6.0 kg
- 2 kinetic energy against velocity for an object of mass 12 kg
- 3 electrical energy transferred against charge transferred across a component with a constant potential difference of 6.0 V
- 4 tension against extension for a system of two identical springs connected in series, each having a spring constant of 6.0 N m^{-1}



- A** 1 and 2
 - B** 1 and 3
 - C** 1 and 4
 - D** 2 and 3
 - E** 2 and 4
 - F** 3 and 4
- 10** A drone travels in a straight line between two points separated by a total distance of 800 m. For the first distance d , it travels at a constant speed of 10 m s^{-1} . For the remainder of the journey, it travels at a constant speed of 25 m s^{-1} . The average speed for the entire journey is 16 m s^{-1} . What is the value of d ?
- A** 200 m
 - B** 250 m
 - C** 300 m
 - D** 400 m
 - E** 500 m
 - F** 600 m

- 11 An air bubble rises through water of constant temperature. A graph of the reciprocal of the bubble's volume, $1/V$, against its depth, h , is a straight line as shown.



What is the atmospheric pressure at the surface of the water?

(Use: density of water, $\rho = 1000 \text{ kg m}^{-3}$; gravitational field strength, $g = 10 \text{ N kg}^{-1}$)

- A $1.0 \times 10^5 \text{ Pa}$
- B $2.0 \times 10^4 \text{ Pa}$
- C $5.0 \times 10^4 \text{ Pa}$
- D $2.0 \times 10^5 \text{ Pa}$
- E $1.0 \times 10^4 \text{ Pa}$
- F $5.0 \times 10^5 \text{ Pa}$

- 12 A resistor of constant resistance is connected to a power supply providing a constant voltage V . A total charge Q passes through the resistor, dissipating a total amount of electrical energy E .

The resistor is then connected to a different power supply providing a constant voltage $2V$.

What is the total charge that must pass through the resistor for the same total energy E to be dissipated?

- | | |
|---|---------------|
| A | $\frac{Q}{4}$ |
| B | $\frac{Q}{2}$ |
| C | Q |
| D | $2Q$ |
| E | $4Q$ |
| F | $\sqrt{2} Q$ |
| G | |

- 13 A student performs a calorimetry experiment using a shiny copper beaker containing hot water. To improve the accuracy of the enthalpy change determination, the student considers several modifications to reduce heat loss to the surroundings.

Which of the following modifications would be **counter-productive** (i.e., would increase the rate of heat loss)?

- A Coating the outer surface of the beaker with a thin layer of dull black paint
- B Polishing the outer surface of the beaker to a mirror finish
- C Placing a plastic lid on top of the beaker
- D Placing the beaker on a cork mat
- E Wrapping the sides of the beaker in mineral wool

- 14** A DC power supply of constant voltage is connected to a fixed resistive load of resistance $2R$ using connecting wires. The total resistance of the connecting wires is initially R . The connecting wires are replaced by new wires of the same material and cross-sectional area, but four times the length. What is the ratio of the power dissipated in the new wires to the power dissipated in the original wires?
- A** 1 : 4
B 1 : 2
C 1 : 1
D 2 : 1
E 4 : 1
- 15** Two solid spheres are made from the same material. Sphere Q has twice the radius of sphere P. Both spheres are heated to the same temperature and allowed to cool in the same environment. The initial rate of fall of temperature of sphere P is R . Which expression gives the initial rate of fall of temperature of sphere Q?
- A** $R/8$
B $R/4$
C $R/2$
D R
E $2R$
F $4R$
G $8R$
- 16** Two identical, inverted test tubes, X and Y, contain trapped air. Tube X has a total mass m_X and tube Y has a total mass m_Y , where $m_X > m_Y$. Both tubes are submerged in water and come to rest, neutrally buoyant, at depths h_X and h_Y respectively. The temperature is constant throughout. Which statement correctly compares the depths h_X and h_Y ?
- A** $h_X < h_Y$
B $h_X > h_Y$
C $h_X = h_Y$
D The relationship depends on the specific ratio m_X/m_Y .
E The relationship depends on the value of atmospheric pressure.
F There is not enough information to determine the relationship.

- 17** A resistor of constant resistance dissipates 36 W when the potential difference across it is 12 V. The potential difference is changed to 8 V. What is the power dissipated by the resistor at this new potential difference?
- A** 16 W
 - B** 24 W
 - C** 36 W
 - D** 54 W
 - E** 81 W
- 18** A mineral sample contains a radioactive parent isotope X and its stable daughter product Y . The ratio of the number of nuclei of X to the number of nuclei of Y is 1 : 3. Assuming that no Y was present initially and that X decays with a half-life of τ , what is the age of the sample?
- A** 1.5τ
 - B** 2τ
 - C** 2.5τ
 - D** 3τ
 - E** 4τ

- 19** An object of constant mass m is in motion. A graph of its kinetic energy against the square of its momentum (p^2) is plotted.

The graph is a straight line passing through the origin with gradient G .

What is the mass m of the object in terms of G ?

- | | |
|----------|----------------|
| A | $\frac{1}{2G}$ |
| B | $\frac{1}{G}$ |
| C | $\frac{G}{2}$ |
| D | $2G$ |
| E | $\frac{2}{G}$ |
| F | G |

- 20** A number of different ions are all travelling at the same speed. They enter a region of uniform magnetic field, which is directed perpendicular to their velocity, and they subsequently travel in circular paths.

The ions are composed of protons (p), neutrons (n), and electrons (e) as listed below. Which ion follows the path with the largest radius?

- A** 20 p, 20 n, 18 e
B 20 p, 22 n, 18 e
C 20 p, 23 n, 18 e
D 20 p, 24 n, 18 e
E 20 p, 26 n, 18 e

- 21** A wave travels from medium P to medium Q. In medium P, the wave speed is 400 m/s and the wavelength is 0.60 m. In medium Q, the wave speed is 1200 m/s.

What is the wavelength of the wave in medium Q?

- A** 0.20 m
- B** 0.60 m
- C** 1.80 m
- D** 5.40 m

- 22** A particle is projected vertically upwards on a planet. In the absence of an atmosphere, the particle would reach a maximum height H_0 .

In the presence of an atmosphere, the particle experiences a constant resistive force of magnitude $\frac{1}{4}$ of its weight.

Which expression gives the maximum height H reached by the particle in the presence of the atmosphere?

- A** $\frac{3}{4}H_0$
- B** $\frac{4}{5}H_0$
- C** H_0
- D** $\frac{5}{4}H_0$
- E** $\frac{4}{3}H_0$

- 23** In an AC circuit, the charge Q stored on a capacitor varies sinusoidally with time. The amplitude of the charge oscillation is Q_0 and the frequency is f .

What is the average magnitude of the current in the circuit over many complete cycles?

- A** 0
- B** fQ_0
- C** $2fQ_0$
- D** $4fQ_0$
- E** $2\pi fQ_0$

- 24** Two ideal transformers, T1 and T2, have their primary coils connected in parallel to the same AC voltage source. The primary coils have an identical number of turns.

The secondary coil of T1 has N_{s1} turns and is connected to a resistor of resistance R . The secondary coil of T2 has N_{s2} turns and is connected to a resistor of resistance $4R$.

The power dissipated in both resistors is the same.

What is the value of the ratio $\frac{N_{s2}}{N_{s1}}$?

- A** $1/2$
 - B** 1
 - C** $\sqrt{2}$
 - D** 2
 - E** 4
- 25** Two cyclists, X and Y, travel along the same straight route. Cyclist X travels at a constant speed of 4.0 m/s and cyclist Y travels at a constant speed of 6.0 m/s . They start at the same time from the same point. Cyclist Y arrives 10 minutes earlier than cyclist X.

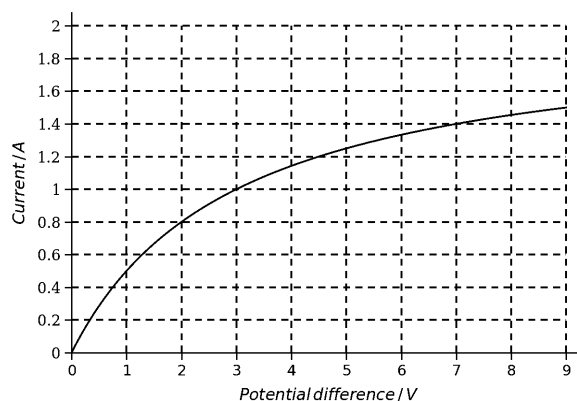
What is the total travel time for cyclist X?

- A** 5.0 minutes
- B** 10 minutes
- C** 15 minutes
- D** 20 minutes
- E** 30 minutes
- F** 40 minutes

- 26** A non-linear electrical component X has the current–potential difference ($I - V$) characteristic shown in the graph below.

Two identical components of type X are connected in parallel with each other. This parallel combination is connected in series with a $3.0\ \Omega$ resistor and an ideal d.c. power supply of electromotive force (EMF) $9.0\ \text{V}$ with negligible internal resistance.

What is the total power dissipated in the entire circuit?



- A** 3.0 W
B 6.0 W
C 9.0 W
D 12 W
E 15 W
F 18 W
G 24 W
H 27 W
- 27** A stationary sound source emits a continuous note of frequency $3300\ \text{Hz}$. A stationary observer is positioned some distance away from the source.
- A wind blows directly from the source to the observer at a constant speed of $30\ \text{m/s}$. The speed of sound in still air is $330\ \text{m/s}$.
- What is the frequency of the sound detected by the observer?
- A** 3000 Hz
B 3025 Hz
C 3300 Hz
D 3600 Hz
E 3630 Hz

Chemistry

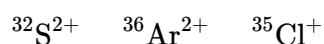
- 1 Equal volumes of $0.100 \text{ mol dm}^{-3} \text{ HCl(aq)}$ (a strong acid) and $0.100 \text{ mol dm}^{-3} \text{ CH}_3\text{COOH(aq)}$ (a weak acid) are investigated in separate experiments at 25°C .

Which of the following statements is/are correct?

- 1 The initial rate of reaction when excess solid calcium carbonate is added is greater with HCl(aq) than with $\text{CH}_3\text{COOH(aq)}$.
- 2 When excess magnesium is added to each acid solution, the total volume of hydrogen gas produced (measured at room temperature and pressure) after both reactions have finished is greater with HCl(aq) than with $\text{CH}_3\text{COOH(aq)}$.
- 3 The volume of $0.0500 \text{ mol dm}^{-3} \text{ Ba(OH)}_2 \text{ (aq)}$ required to completely neutralise the acid solution is the same for both acids.

- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 2 Three positive ions are detected in a mass spectrometer:



Which of the following statements is/are correct?

- 1 $^{36}\text{Ar}^{2+}$ and $^{35}\text{Cl}^{+}$ have the same electron configuration.
- 2 $^{36}\text{Ar}^{2+}$ has more neutrons than $^{35}\text{Cl}^{+}$.
- 3 $^{32}\text{S}^{2+}$ has the lowest mass-to-charge ratio (m/z) of the three ions.

- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

3 Four metals W, X, Y and Z have the following properties.

- W displaces X from aqueous X sulfate.
- X reacts with dilute hydrochloric acid, but Y does not.
- Z is extracted from its molten compound by electrolysis.
- W can be extracted from its oxide by reduction with carbon.

Which order shows decreasing reactivity?

- A** $Z > W > X > Y$
B $Z > X > W > Y$
C $W > Z > X > Y$
D $Z > W > Y > X$
E $W > X > Z > Y$

4 In each of the following four procedures, an excess of an aqueous reagent is added to 1.0 dm^3 of a 1.0 mol dm^{-3} aqueous solution of a salt:

- excess aqueous silver nitrate added to magnesium chloride solution
- excess aqueous barium chloride added to aluminium sulfate solution
- excess aqueous sodium hydroxide added to copper(II) sulfate solution
- excess aqueous sodium chloride added to potassium nitrate solution

Which row in the following table identifies the procedures that form a precipitate and produce the largest, and the smallest, theoretical mass of precipitate?

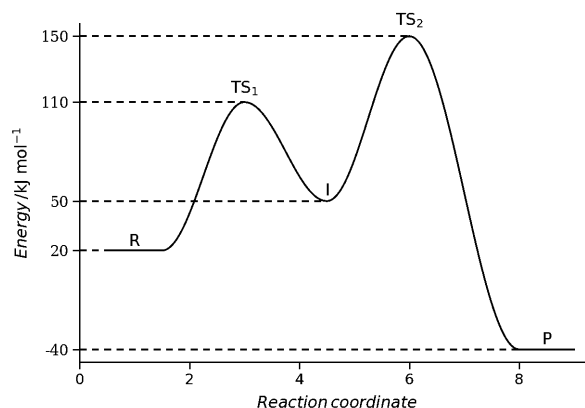
(M_r values: $\text{AgCl} = 143.5$; $\text{BaSO}_4 = 233$; $\text{Cu}(\text{OH})_2 = 98$; $\text{KCl} = 74.5$; $\text{NaNO}_3 = 85$)

	Procedure that produces the largest mass of precipitate	Procedure that produces the smallest mass of precipitate
A	AgNO_3 and MgCl_2	NaOH and CuSO_4
B	AgNO_3 and MgCl_2	NaCl and KNO_3
C	BaCl_2 and $\text{Al}_2(\text{SO}_4)_3$	AgNO_3 and MgCl_2
D	BaCl_2 and $\text{Al}_2(\text{SO}_4)_3$	NaOH and CuSO_4
E	BaCl_2 and $\text{Al}_2(\text{SO}_4)_3$	NaCl and KNO_3
F	NaOH and CuSO_4	BaCl_2 and $\text{Al}_2(\text{SO}_4)_3$

- 5 The reaction profile for a two-step uncatalysed reaction $R \rightarrow I \rightarrow P$ is shown below.

In the presence of a catalyst, the activation energy of the forward rate-determining step is reduced by 30 kJ mol^{-1} , while the activation energy of the other step is unaffected.

What is the activation energy of the rate-determining step for the reverse catalysed reaction $P \rightarrow R$?



- A 60 kJ mol^{-1}
 B 70 kJ mol^{-1}
 C 100 kJ mol^{-1}
 D 130 kJ mol^{-1}
 E 160 kJ mol^{-1}
 F 190 kJ mol^{-1}
- 6 Which of the following reactions involve(s) oxidation?
- 1 $2\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$
 2 $\text{MnO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 5\text{e}^- \rightarrow \text{Mn}^{2+}(\text{aq}) + 4\text{H}_2\text{O}(\text{l})$
 3 $2\text{CrO}_4^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) \rightarrow \text{Cr}_2\text{O}_7^{2-}(\text{aq}) + \text{H}_2\text{O}(\text{l})$
 4 $\text{SO}_2(\text{g}) + 2\text{H}_2\text{S}(\text{g}) \rightarrow 3\text{S}(\text{s}) + 2\text{H}_2\text{O}(\text{l})$
- A 1 and 2 only
 B 1 and 3 only
 C 1 and 4 only
 D 2 and 3 only
 E 2 and 4 only
 F 3 and 4 only
 G 1, 2 and 4 only
 H 1, 3 and 4 only

- 7 The table shows some physical properties of four pure solid substances, W, X, Y, and Z.

Substance	Melting point / °C	Electrical conductivity as a solid	Electrical conductivity when molten	Electrical conductivity of aqueous solution
W	772	poor	good	good
X	1650	poor	poor	insoluble
Y	122	poor	poor	good
Z	146	poor	poor	poor

Consider the following statements:

1. The electrical conductivity of molten W is due to the movement of delocalised electrons.
2. Melting substance X involves breaking strong covalent bonds.
3. When substance Y is added to water, mobile ions are produced.

Which of the statements is/are correct?

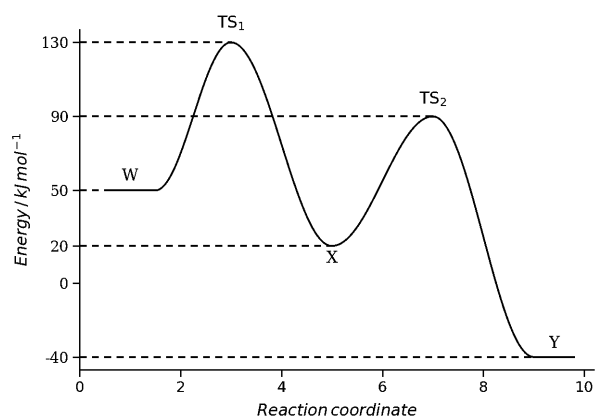
- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

8 The reaction energy profile for the two-step reaction $W \rightarrow X \rightarrow Y$ is shown below.

Which row in the table gives the correct values for:

- the overall enthalpy change (ΔH) for the forward reaction $W \rightarrow Y$;
- the activation energy (E_a) of the rate-determining step for the forward reaction;
- the activation energy (E_a) of the rate-determining step for the reverse reaction $Y \rightarrow W$?

	ΔH (overall forward) / $\text{kJ}\cdot\text{mol}^{-1}$	E_a (forward RDS) / $\text{kJ}\cdot\text{mol}^{-1}$	E_a (reverse RDS) / $\text{kJ}\cdot\text{mol}^{-1}$
A	-90	+80	+130
B	-90	+80	+170
C	-90	+70	+110
D	-90	+130	+130
E	+90	+80	+130
F	+90	+80	+170
G	+90	+70	+110
H	+90	+130	+170



- 9 In this question, all aqueous solutions have the same concentration in mol dm^{-3} .

Three separate experiments were carried out at room temperature by mixing equal volumes of the following solutions:

experiment 1: $\text{MgSO}_4(\text{aq})$ and $\text{Ba}(\text{OH})_2(\text{aq})$

experiment 2: $\text{Na}_2\text{SO}_4(\text{aq})$ and $\text{Ba}(\text{OH})_2(\text{aq})$

experiment 3: $\text{K}_2\text{SO}_4(\text{aq})$ and $\text{Ba}(\text{OH})_2(\text{aq})$

For each experiment, the resulting mixture was filtered, and the solid residue was washed, dried, and weighed.

Which of the following statements is/are correct?

1 Experiment 1 produces a greater mass of dry solid than Experiment 2.

2 Experiment 3 produces a greater mass of dry solid than Experiment 2.

3 After removing the solid, the filtrate from Experiment 1 has a lower concentration of dissolved ions than the filtrate from Experiment 2.

- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

10 Three separate aqueous solutions are electrolysed using inert electrodes:

- Solution 1: concentrated NaCl(aq)
- Solution 2: dilute H₂SO₄(aq)
- Solution 3: concentrated CuSO₄(aq)

The same quantity of electrical charge is passed through each solution. Assume that all gases evolved behave ideally and are completely insoluble in water.

Which of the following statements is/are correct?

1 The volume of gas evolved at the cathode in Solution 1 is equal to the volume of gas evolved at the cathode in Solution 2 (measured at the same temperature and pressure).

2 The volume of gas evolved at the anode in Solution 2 is twice the volume of gas evolved at the cathode in Solution 2.

3 As electrolysis proceeds, the pH of Solution 1 increases and the pH of Solution 3 decreases.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

11 The equation summarises the oxidation of iodine by concentrated nitric acid:



What values of w and y are needed to balance the equation?

	w	y
A	6	2
B	8	3
C	10	4
D	10	5
E	12	4

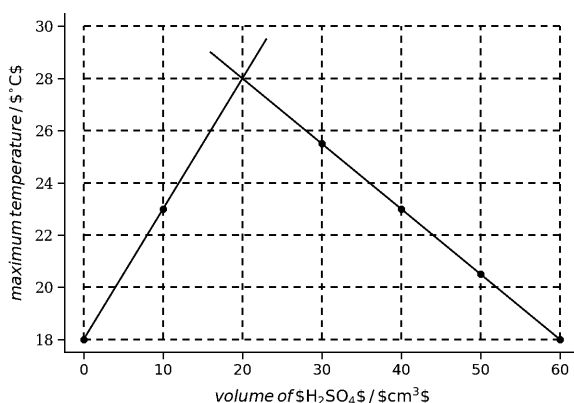
- 12** In a series of experiments to determine the molar enthalpy change of neutralisation, various volumes of $1.0 \text{ mol dm}^{-3} \text{ H}_2\text{SO}_4 (\text{aq})$ and $1.0 \text{ mol dm}^{-3} \text{ NaOH (aq)}$ were mixed in an insulated polystyrene cup such that the total volume was always 60 cm^3 .

The initial temperature of both solutions was 18.0°C . The maximum temperature reached for each mixture was recorded and the results plotted on the graph shown.

Two straight lines of best fit were drawn and extrapolated to intersect at the maximum.

What is the molar enthalpy change of neutralisation (per mole of water formed) for this reaction, in kJ mol^{-1} ?

(Assume that the specific heat capacity of all solutions is $4.0 \text{ J g}^{-1}^\circ\text{C}^{-1}$, the density of dilute solutions is 1.0 g cm^{-3} , and all heat transferred stays in the solution.)

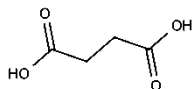


- A** -2.4 kJ mol^{-1}
- B** -30 kJ mol^{-1}
- C** -40 kJ mol^{-1}
- D** -60 kJ mol^{-1}
- E** -120 kJ mol^{-1}
- F** $-2400 \text{ kJ mol}^{-1}$

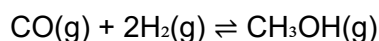
- 13** A sample of 1.18 g of the dicarboxylic acid shown below is reacted with an excess of aqueous sodium hydrogencarbonate ($\text{NaHCO}_3(\text{aq})$) at room temperature and pressure.

Assuming that 1 mol of gas occupies 24.0 dm^3 under these conditions, what is the volume of gas collected?

(A_r values: H = 1.0, C = 12.0, O = 16.0, Na = 23.0)



- A** 0.24 cm^3
B 0.48 cm^3
C 120 cm^3
D 240 cm^3
E 480 cm^3
F 960 cm^3
- 14** Methanol can be manufactured industrially by the reversible reaction of carbon monoxide with hydrogen in the presence of a catalyst:



In a closed reaction vessel at constant temperature and pressure, a mixture containing 8.00 mol of $\text{CO}(\text{g})$ and 20.0 mol of $\text{H}_2(\text{g})$ is allowed to reach equilibrium. At equilibrium, 75.0% of the CO has been converted to $\text{CH}_3\text{OH}(\text{g})$.

What is the total volume of the resulting equilibrium gas mixture?

(Assume that under these conditions, one mole of gas occupies a volume of 45.0 dm^3 .)

- A** 270 dm^3
B 450 dm^3
C 540 dm^3
D 630 dm^3
E 720 dm^3
F 990 dm^3
G 1260 dm^3

- 15** The table shows the melting points and electrical conductivities of four pure substances, P, Q, R, and S, at standard atmospheric pressure:

Substance	Melting point / °C	Electrical conductivity of solid	Electrical conductivity of liquid
P	−78	non-conductor	non-conductor
Q	770	non-conductor	good conductor
R	1420	non-conductor	non-conductor
S	1538	good conductor	good conductor

Which of the following statements is/are correct?

- 1 Melting substance P requires breaking strong covalent bonds between atoms.
- 2 The high melting point of substance R is due to the breaking of strong covalent bonds in a giant lattice.
- 3 In molten Q, electrical conduction is due to mobile ions, whereas in solid S, it is due to delocalised electrons.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 16** Element Q is in Group 17, Period 4 of the Periodic Table. It exists naturally as a mixture of two isotopes with mass numbers 79 and 81. The relative atomic mass of element Q is 79.9.

Which of the following statements is/are correct about element Q?

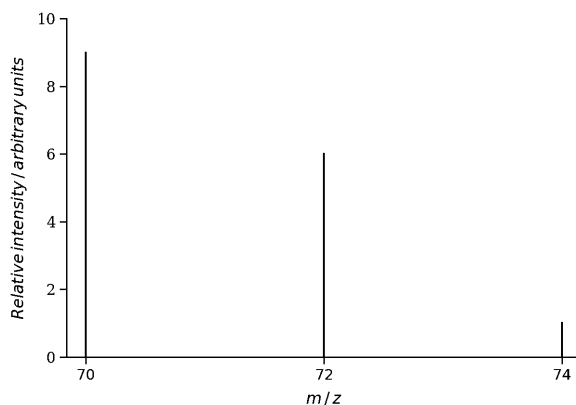
1. 55% of the atoms in a sample of element Q have a mass number of 79.
2. Aqueous element Q oxidises chloride ions to chlorine.
3. Element Q reacts with calcium to form a compound with the empirical formula CaQ_2 that conducts electricity when molten.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 17 The mass spectrum of the molecular ion region for a sample of a pure diatomic gas, X_2 , is shown below.

Element X reacts with phosphorus to form the trihalide PX_3 . Naturally occurring phosphorus consists exclusively of the single stable isotope ^{31}P .

How many peaks appear in the molecular ion region of the mass spectrum of PX_3 , and what is the ratio of the height of the lightest molecular ion peak to that of the heaviest molecular ion peak?



- A 3 peaks; ratio 9 : 1
- B 3 peaks; ratio 27 : 1
- C 4 peaks; ratio 3 : 1
- D 4 peaks; ratio 9 : 1
- E 4 peaks; ratio 27 : 1
- F 4 peaks; ratio 81 : 1
- G 6 peaks; ratio 27 : 1
- H 8 peaks; ratio 27 : 1

- 18** A solid element **M** burns in air with an intense, bright white flame to form a single white solid oxide.

This oxide reacts completely with a dilute aqueous solution of acid **A** to form a colourless solution containing a single salt.

When aqueous barium chloride is added to a sample of acid **A**, followed by dilute hydrochloric acid, a dense white precipitate remains.

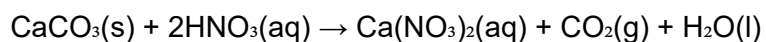
When aqueous sodium hydroxide is added dropwise to the solution of the salt formed from the oxide of **M** and acid **A**, a white precipitate forms which does not dissolve in excess sodium hydroxide.

Acid **A** also reacts with an insoluble green solid carbonate **Q** to produce three products, one of which is a blue solution.

Which row in the table correctly identifies **M**, **A**, and **Q**?

	Element M	Acid A	Carbonate Q
A	Ca	HCl	CuCO ₃
B	Ca	H ₂ SO ₄	FeCO ₃
C	Mg	HCl	CaCO ₃
D	Mg	HNO ₃	CuCO ₃
E	Mg	H ₂ SO ₄	CaCO ₃
F	Mg	H ₂ SO ₄	CuCO ₃
G	Zn	HNO ₃	FeCO ₃
H	Zn	H ₂ SO ₄	CuCO ₃

- 19** An excess of dilute nitric acid was added to a sample of solid calcium carbonate in an open conical flask placed on an electronic balance:



The total mass of the flask and its contents was recorded at the start ($t = 0 \text{ s}$) and at 10 s intervals thereafter.

The reaction was complete (stopped) exactly at $t = 40 \text{ s}$.

Which row in the table could represent the readings displayed on the electronic balance during this experiment?

	0 s	10 s	20 s	30 s	40 s	50 s
A	200.0 g	197.6 g	196.4 g	195.8 g	195.6 g	195.6 g
B	200.0 g	197.6 g	196.4 g	195.8 g	195.6 g	195.5 g
C	200.0 g	197.4 g	196.0 g	195.6 g	195.6 g	195.6 g
D	200.0 g	198.9 g	197.8 g	196.7 g	195.6 g	195.6 g
E	200.0 g	199.8 g	199.2 g	198.0 g	195.6 g	195.6 g
F	200.0 g	202.4 g	203.6 g	204.2 g	204.4 g	204.4 g

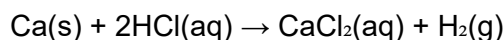
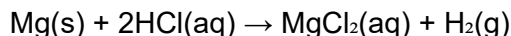
- 20** Three open beakers, each containing an excess of dilute hydrochloric acid, HCl(aq) , are placed on separate digital balances.

The initial mass reading on each balance is recorded.

Equal masses (5.00 g) of three pure solids are added separately to the beakers:

- Beaker 1: magnesium metal, Mg(s)
- Beaker 2: calcium metal, Ca(s)
- Beaker 3: magnesium carbonate, $\text{MgCO}_3\text{(s)}$

In each beaker, a reaction occurs and the gas produced escapes freely into the atmosphere:



Assume that all reactions go to completion, no liquid splashes out, and evaporation of water is negligible.

[A_r values: $\text{H} = 1.0$; $\text{C} = 12.0$; $\text{O} = 16.0$; $\text{Mg} = 24.3$; $\text{Ca} = 40.1$]

Which of the following statements is/are correct?

- 1 The final balance reading for each beaker is greater than its initial reading before the solid was added.
- 2 Beaker 1 produces the greatest volume of gas, measured at room temperature and pressure.
- 3 Beaker 1 loses the greatest mass of gas to the surroundings.

- A** none
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

21 Which of the following species have all of their atoms lying in the same plane?



A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

22 The hydrolysis of one mole of triglyceride into glycerol and fatty acids has an enthalpy change of $\Delta H = +15 \text{ kJ mol}^{-1}$. The table gives the bond energies for the bonds involved, except for the ester C-O bond.

Bond	Bond Energy / kJ mol^{-1}
O-H (water)	464
O-H (glycerol)	455
C-O (fatty acid)	358

Assume that the hydrolysis of each ester linkage involves breaking one ester C-O bond and one water O-H bond, and forming one fatty acid C-O bond and one glycerol O-H bond.

What is the bond energy of the ester C-O bond?

A 344 kJ mol^{-1}

B 354 kJ mol^{-1}

C 362 kJ mol^{-1}

D 363 kJ mol^{-1}

E 364 kJ mol^{-1}

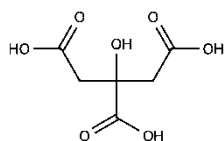
- 23** A 1.60 g sample containing compound **X** and an unreactive impurity was added to an excess of dilute aqueous sodium hydrogencarbonate, $\text{NaHCO}_3(\text{aq})$.

The structure of compound **X** is shown below.

The volume of carbon dioxide gas collected at room temperature and pressure was 360 cm^3 .

What is the mass of the impurity in the sample?

(A_r : H = 1.0, C = 12.0, O = 16.0; molar gas volume at room temperature and pressure = $24\,000 \text{ cm}^3 \text{ mol}^{-1}$)



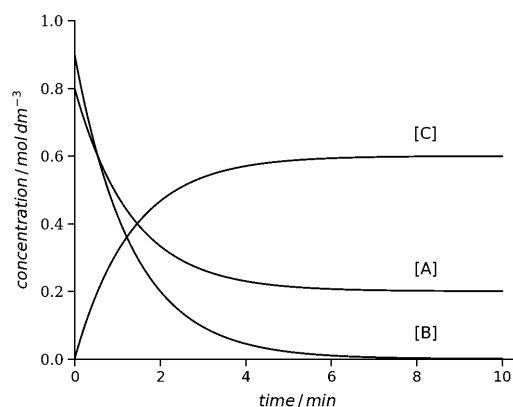
- A** 0.16 g
- B** 0.64 g
- C** 0.72 g
- D** 0.88 g
- E** 0.96 g
- F** 1.44 g

- 24** Substances A and B react in a closed container of fixed volume to form substance C. The reaction goes to completion.

The graph shows how the concentrations of A, B, and C change over time.

Which row in the table correctly gives the balanced equation for the reaction, the limiting reactant, and the percentage of the initial amount of the excess reactant that has reacted?

	Balanced equation	Limiting reactant	Percentage of excess reactant reacted
A	$2A + 3B \rightarrow 2C$	B	75%
B	$2A + 3B \rightarrow 2C$	B	25%
C	$2A + 3B \rightarrow 2C$	A	75%
D	$3A + 2B \rightarrow 2C$	B	75%
E	$3A + 2B \rightarrow 2C$	A	25%
F	$4A + 3B \rightarrow 2C$	B	75%
G	$4A + 3B \rightarrow 2C$	A	25%
H	$A + 3B \rightarrow 2C$	B	33%



- 25** Gas-liquid chromatography with a non-polar stationary phase was used to analyse three hydrocarbon mixtures labelled Sample X, Sample Y, and Sample Z.

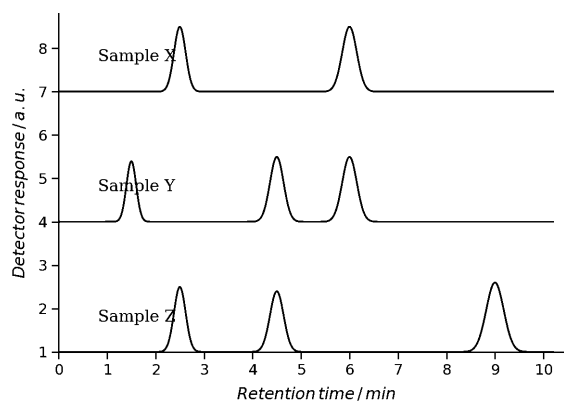
The table gives the retention times of five pure reference hydrocarbons obtained under the same experimental conditions:

Hydrocarbon	Retention time / min
methylpropane	1.5
butane	2.5
2-methylbutane	4.5
pentane	6.0
hexane	9.0

The chromatogram obtained for the three samples is shown.

Which of the following statements is/are correct?

- 1 Sample X contains only unbranched alkanes.
- 2 Butane is present in all three samples.
- 3 Both Sample Y and Sample Z contain at least one hydrocarbon with the molecular formula C_5H_{12} .
- 4 Hexane is the most volatile of the five reference hydrocarbons.



- A** 1 only
- B** 3 only
- C** 1 and 3 only
- D** 1 and 4 only
- E** 2 and 3 only
- F** 2 and 4 only
- G** 1, 3 and 4 only
- H** 1, 2, 3 and 4

- 26** An aqueous solution of barium hydroxide, $\text{Ba}(\text{OH})_2(\text{aq})$, is mixed in an equimolar ratio with dilute sulfuric acid, $\text{H}_2\text{SO}_4(\text{aq})$. A white precipitate of barium sulfate, $\text{BaSO}_4(\text{s})$, forms alongside liquid water, $\text{H}_2\text{O}(\text{l})$, causing the electrical conductivity of the mixture to drop to near zero.

Which of the following represents the correct balanced ionic equation for this reaction?

- A** $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$
 - B** $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$
 - C** $\text{Ba}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + 2\text{H}_2\text{O}(\text{l})$
 - D** $\text{Ba}(\text{OH})_2(\text{aq}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + 2\text{H}_2\text{O}(\text{l})$
 - E** $\text{Ba}^{2+}(\text{aq}) + \text{OH}^-(\text{aq}) + \text{H}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + \text{H}_2\text{O}(\text{l})$
 - F** $\text{Ba}(\text{OH})_2(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + 2\text{H}_2\text{O}(\text{l})$
- 27** The monatomic ions X^{2+} , Y^- and Z^{3-} are isoelectronic (they contain the same number of electrons).

Which of the following gives the elements X, Y and Z in order of increasing atomic number (lowest first)?

- A** X, Y, Z
- B** Z, Y, X
- C** Y, X, Z
- D** Z, X, Y

Biology

- 1 In an agricultural region, soil was repeatedly treated with a chemical nematicide over a period of ten years to control pest worms. During this time, the population of a native worm, *Species Y*, declined continuously and eventually died out completely. In contrast, an invasive worm, *Species X*, suffered an initial decline in numbers but subsequently recovered and thrived despite continued application of the nematicide.

A student made the following statements to explain these observations:

- 1 The application of the nematicide acted as a selective pressure that increased the frequency of resistance alleles in the population of *Species X*.
- 2 The nematicide induced directed mutations in *Species X* to allow individuals to adapt to the chemical.
- 3 *Species Y* became locally extinct because none of its individuals possessed alleles conferring sufficient tolerance to survive the nematicide.

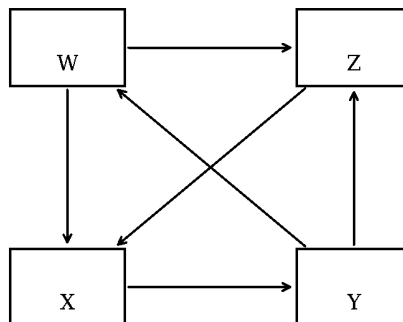
Which of the student's statements is/are correct?

- A 1 only
- B 2 only
- C 3 only
- D 1 and 2 only
- E 1 and 3 only
- F 2 and 3 only
- G 1, 2 and 3

- 2 The diagram shows part of the nitrogen cycle in an agricultural ecosystem containing legume crops.

	Ammonium ions in soil	Atmospheric nitrogen gas	Nitrate ions in soil	Nitrogen in plant biomass
A	W	X	Z	Y
B	X	W	Y	Z
C	X	Y	W	Z
D	Y	W	X	Z
E	Z	W	X	Y
F	X	Z	Y	W
G	Y	X	W	Z
H	Z	X	Y	W

Which row correctly identifies boxes W, X, Y, and Z?



- 3 The table shows three ecological characteristics for six freshwater fish species (A–F) inhabiting a river system subject to severe seasonal drought and habitat fragmentation:

species	dietary niche	dispersal ability	generation time
A	generalist	high	short
B	specialist	low	short
C	specialist	high	long
D	specialist	low	long
E	generalist	low	long
F	generalist	high	long

Which species is most vulnerable to local extinction as environmental conditions become persistently harsher?

- 4 A student analysed a section of a gene identified in four different organisms. The gene codes for a functional polypeptide involved in ATP synthesis.

A 15-base-pair section of the coding (non-template) DNA strand is shown in the table below. The remainder of the gene (not shown) is identical in sequence in all four organisms.

Organism	DNA sequence (5' to 3')
human	ATG CTT GGA GTC AAA
baker's yeast	ATG CTT GGT GTC AAA
pea plant	ATG CTC GGT GTT AAA
cyanobacterium	ATG AAC CGT TCG AAA

The student made the following conclusions:

- 1 This gene could code directly for a polypeptide subunit of ATP synthase, but cannot code directly for ATP.
- 2 In the cyanobacterium, transcription of this gene and translation of its mRNA can occur simultaneously.
- 3 Based on the section shown, the DNA sequence of the pea plant is more similar to that of the cyanobacterium than to that of baker's yeast.

Which of these conclusions is/are correct?

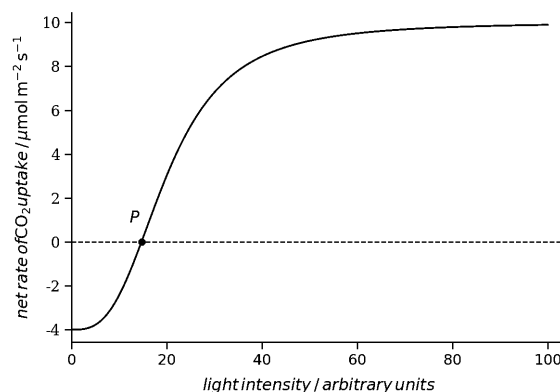
- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 5 A population of fish in a lake consists of two distinct phenotypes, Morph A and Morph B. Their characteristics are summarized below:
- | Trait | Morph A | Morph B |
|----------------------------|--------------------------|---------------------------------|
| Appearance | Cryptic (camouflaged) | Conspicuous (brightly coloured) |
| Competitive Ability | Weak competitor for food | Strong competitor for food |
| Metabolic Rate | Low | Low |
- Historically, the lake water was clear, and the fish were preyed upon by a visual predator. Recently, agricultural runoff has caused the water to become permanently turbid (cloudy), significantly reducing visibility. Simultaneously, the turbidity has reduced light penetration, causing a 50% decrease in the aquatic plants that serve as the food source for the fish. Which prediction for the long-term evolutionary trajectory of this population is most supported by the information provided?
- A The relative frequency of Morph B will increase because the penalty for being conspicuous is reduced while the advantage of being a strong competitor becomes more critical.
 - B The relative frequency of Morph A will increase because camouflage becomes even more effective in turbid water, enhancing survival.
 - C The relative frequency of Morph A will increase because the reduction in food availability will disadvantage the conspicuous Morph B more than Morph A.
 - D The relative frequency of Morph B will decrease because the predator will switch to non-visual hunting methods, negating the effect of turbidity.
 - E The relative frequencies will remain unchanged because the reduction in predation pressure cancels out the reduction in food availability.

- 6 The graph shows the effect of light intensity on the net rate of carbon dioxide (CO₂) uptake by a leaf at a constant temperature and atmospheric CO₂ concentration.

Which of the following statements is/are correct?

- 1 At a light intensity of 0 arbitrary units, the rate of CO₂ release by cellular respiration is $4 \mu\text{mol m}^{-2} \text{s}^{-1}$.
- 2 At point P, the rate of the light-dependent reactions of photosynthesis is zero.
- 3 Assuming the rate of cellular respiration remains constant across all light intensities, the rate of gross CO₂ fixation at a light intensity of 80 arbitrary units is $14 \mu\text{mol m}^{-2} \text{s}^{-1}$.

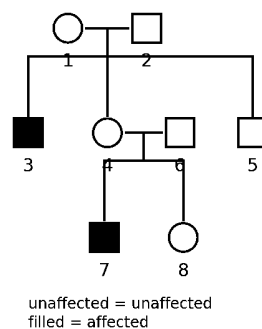


- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 7 The pedigree diagram shows the inheritance of a genetic condition in a family. The condition is controlled by a single gene with two alleles. Assume complete penetrance and that no new mutations occur.

Which of the following statements is/are correct?

- 1 The condition cannot be caused by a dominant allele.
- 2 If the condition is sex-linked (X-linked) recessive, the probability that individual 8 is a carrier of the allele is 50%.
- 3 If the condition is autosomal recessive, the probability that individual 4 is a carrier of the allele is $\frac{2}{3}$.

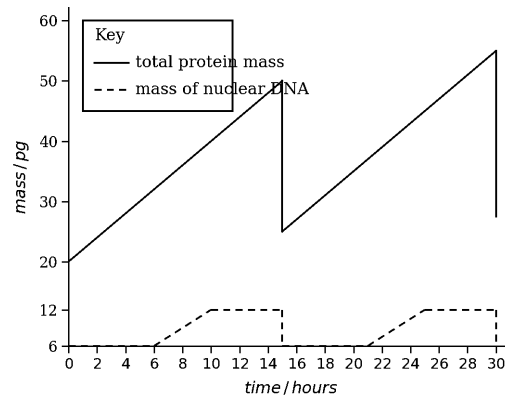


- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 8 The graph shows the changes in the mass of nuclear DNA and the total protein mass of an individual dividing animal cell over a period of 30 hours.

Which of the following statements is/are correct?

- 1 Between 10 and 14 hours, the cell contains twice as many chromatids as it does between 0 and 6 hours.
- 2 The duration of the S phase is greater than the duration of the G_1 phase.
- 3 If an inhibitor of mitotic spindle formation is added to the cell at 8 hours, the mass of nuclear DNA in the cell at 18 hours will be 12.0 pg.

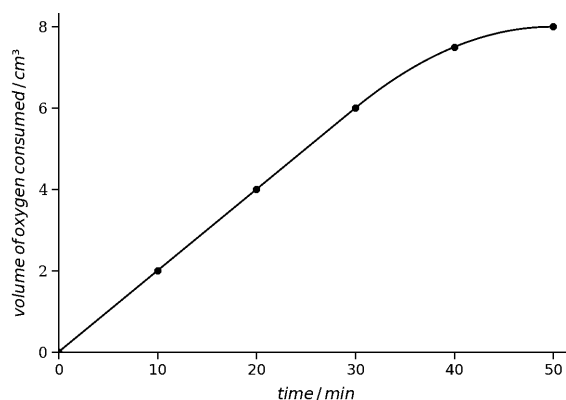


- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 9 A simple respirometer containing potassium hydroxide solution (KOH) was used to measure the aerobic respiration of germinating seeds in a sealed chamber at 20 °C. The cumulative volume of oxygen consumed was recorded over a period of 50 minutes, and the results are plotted on the graph.

Which of the following statements is/are correct?

- 1 The initial rate of oxygen consumption by the seeds was $12.0 \text{ cm}^3 \text{ h}^{-1}$.
- 2 If the experiment were repeated with seeds metabolising only lipids ($RQ = 0.7$) without KOH, the total volume of gas in the respirometer chamber would increase over time.
- 3 The reduction in the rate of oxygen consumption after 30 minutes could be due to oxygen concentration becoming a limiting factor.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

10 In birds, sex is determined by the ZW sex-chromosome system:

- * males are homogametic (ZZ)
- * females are heterogametic (ZW)

The W chromosome does not carry an allele for the feather pattern gene.

Feather barring in chickens is controlled by a gene on the Z chromosome:

- * the dominant allele, Z^B , produces barred feathers
- * the recessive allele, Z^b , produces non-barred feathers

A breeder carries out two separate crosses:

- * **Cross 1:** a heterozygous barred male ($Z^B Z^b$) is mated with a non-barred female ($Z^b W$)
- * **Cross 2:** a non-barred male ($Z^b Z^b$) is mated with a barred female ($Z^B W$)

What is the expected proportion of all offspring that are non-barred females in Cross 1, and what is the expected proportion of all offspring that are non-barred females in Cross 2?

	Cross 1	Cross 2
A	0.25	0.00
B	0.25	0.25
C	0.25	0.50
D	0.50	0.25
E	0.50	0.50
F	0.50	1.00

- 11 Four different cell types (**P**, **Q**, **R**, and **S**) were analysed for the presence (+) or absence (–) of five cellular structures. The results are shown in the table below:

Cell type	Plasma membrane	Nucleus	Mitochondria	Ribosomes	Cell wall
P	+	+	+	+	+
Q	+	–	–	+	+
R	+	+	+	+	–
S	+	–	–	–	–

Which of the following statements is/are correct?

1 Cell type **P** could be a fungal cell or a plant leaf cell.

2 Cell type **Q** cannot synthesise proteins or generate ATP by cellular respiration because it lacks membrane-bound organelles.

3 Cell type **S** could be a mature human red blood cell (erythrocyte).

- A** 1 only
- B** 2 only
- C** 3 only
- D** 1 and 2 only
- E** 1 and 3 only
- F** 2 and 3 only
- G** 1, 2 and 3
- H** none of them

- 12** Haemophilia A is an X-linked recessive genetic condition in humans caused by a mutation in the gene for clotting Factor VIII. Individuals with this condition cannot produce functional Factor VIII protein, which is normally synthesised by liver cells and secreted into the bloodstream.

In a clinical trial, adult male patients with Haemophilia A receive a gene therapy treatment. A harmless viral vector carrying a functional human Factor VIII gene is injected into the bloodstream. The vector delivers the gene into the nuclei of the patient's liver cells, where it is expressed.

Which of the following statements correctly describe(s) the consequences of this gene therapy in a successfully treated patient?

- 1 The patient's liver cells are able to transcribe and translate the functional Factor VIII gene.
- 2 The patient cannot pass the mutant Factor VIII allele on to any of his future biological daughters.
- 3 The total nucleotide sequence of the DNA inside the patient's liver cells has been altered.

- A** 1 only
- B** 2 only
- C** 3 only
- D** 1 and 2 only
- E** 1 and 3 only
- F** 2 and 3 only
- G** 1, 2 and 3

- 13** A student investigated stomatal density on the lower epidermis of a dicotyledonous leaf.

A transparent peel of the lower epidermis was prepared and examined under a light microscope. The circular field of view of the microscope at high magnification had an area of 0.20 mm^2 . The student counted the number of stomata in six randomly selected fields of view:

Field of view	Number of stomata
1	28
2	34
3	31
4	29
5	35
6	35

The total surface area of the lower surface of the leaf was measured to be 15.0 cm^2 .

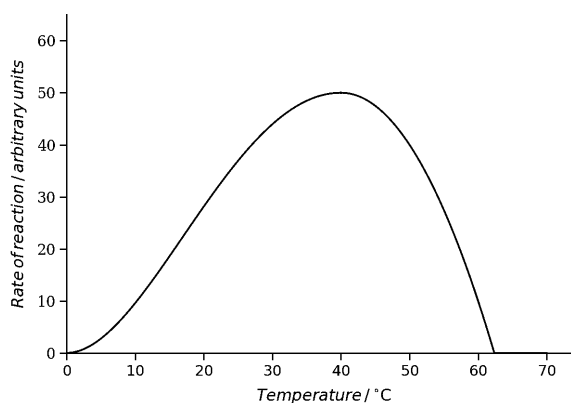
What is the estimated total number of stomata on the lower surface of this leaf?

- A** 2.4×10^3
- B** 2.4×10^4
- C** 1.2×10^5
- D** 2.4×10^5
- E** 4.8×10^5
- F** 1.44×10^6
- G** 2.4×10^6

- 14 The graph shows the effect of temperature on the rate of an enzyme-catalysed reaction.

	Percentage decrease in rate between 40 °C and 60 °C	Explanation for the decrease in rate
A	20%	Increased thermal vibrations break bonds maintaining the tertiary structure, altering the active site
B	20%	High temperature kills the enzyme molecules, stopping their biological activity
C	40%	Increased thermal vibrations break bonds maintaining the tertiary structure, altering the active site
D	80%	Increased thermal vibrations break bonds maintaining the tertiary structure, altering the active site
E	80%	High temperature kills the enzyme molecules, stopping their biological activity
F	80%	The activation energy required for the uncatalysed reaction increases at high temperatures
G	400%	Increased thermal vibrations break bonds maintaining the tertiary structure, altering the active site
H	400%	High temperature kills the enzyme molecules, stopping their biological activity

Which row correctly calculates the percentage decrease in the rate of reaction between 40°C and 60°C and gives the correct biological explanation for this decrease?



- 15** Ecologists surveyed two distinct habitats, Zone 1 and Zone 2, to compare the distribution of two plant species, X and Y. In each zone, 20 randomly placed quadrats were surveyed. The table below shows the number of quadrats in which each species was present and the total number of individuals counted for that species in that zone.

Zone	Species	Number of quadrats containing species	Total number of individuals
1	X	16	32
1	Y	5	120
2	X	8	160
2	Y	18	45

Which of the following conclusions is supported by the data?

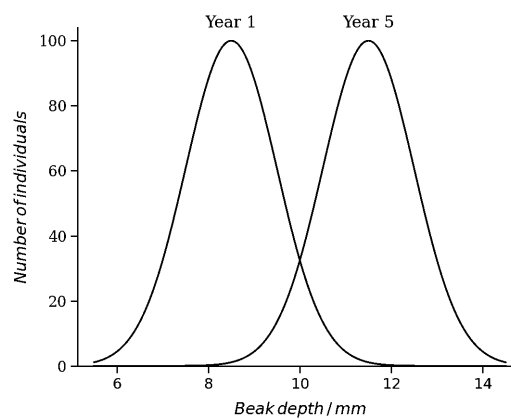
- A** In Zone 1, Species X has a higher frequency of occurrence than Species Y, despite having a lower total abundance.
- B** In Zone 2, Species X has a higher frequency of occurrence than Species Y.
- C** The frequency of occurrence of Species X in Zone 1 is 40%.
- D** Species Y demonstrates a more clumped distribution in Zone 2 than in Zone 1.
- E** The mean density of Species Y in Zone 1 is 0.25 individuals per quadrat.

- 16** A population of a seed-eating bird species lives on an isolated island. A prolonged environmental change resulted in a severe shortage of small, soft seeds, leaving predominantly large, tough-shelled seeds.

The graph shows the distribution of beak depth in the bird population in Year 1 (before the environmental change) and in Year 5 (after four generations under the new conditions).

Which of the following statements is/are correct?

- 1 The environmental change stimulated specific mutations in the birds' DNA to increase their beak depth.
- 2 The change in the population's beak depth distribution is an example of directional selection.
- 3 For this evolutionary change to occur across generations, variation in beak depth must have a genetic (heritable) basis.



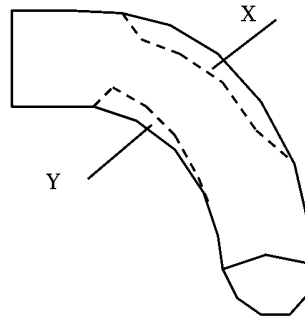
- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 17** The diagram shows the tip of a plant root that was placed horizontally in the dark, with two regions in the elongation zone labelled X and Y.

A student proposed the following statements to explain the growth and bending of this root:

1. Cells in region X are longer than cells in region Y.
2. The concentration of auxin is higher in region Y than in region X.
3. The downward bending occurs because the accumulation of auxin at Y inhibits cell elongation.

Which of the student's statements is/are correct?



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 18** The table below shows the net rate of carbon dioxide uptake by a plant leaf measured at varying light intensities. The temperature and carbon dioxide concentration were maintained at constant levels throughout the experiment.

Light Intensity (arbitrary units)	Net CO ₂ Uptake (mg dm ⁻² h ⁻¹)
0	-2.0
10	0.0
30	5.5
60	9.8
90	10.5
120	10.5

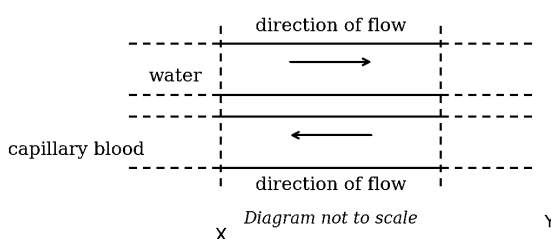
Which of the following conclusions is best supported by the data?

- A** At a light intensity of 10 arbitrary units, the rate of photosynthesis is zero.
- B** Between light intensities of 30 and 60 arbitrary units, the concentration of carbon dioxide is the primary limiting factor.
- C** The increase in net CO₂ uptake from 0 to 10 arbitrary units is primarily due to a decrease in the rate of respiration.
- D** At a light intensity of 120 arbitrary units, the rate of photosynthesis is limited by a factor other than light intensity.
- E** The maximum rate of gross photosynthesis achieved in this experiment is 10.5 mg dm⁻² h⁻¹.
- 19** The diagram shows a simplified model of gas exchange across a secondary lamella in the gill of a bony fish.

Water flows across the lamella from position X to position Y. Capillary blood flows in the opposite direction, from position Y to position X.

Which of the following statements is/are correct?

- 1 Moving along the exchange surface from position X to position Y, the partial pressure of oxygen in the capillary blood decreases.
- 2 The partial pressure of oxygen in the blood leaving at position X can exceed the partial pressure of oxygen in the water leaving at position Y.
- 3 At position Y, net diffusion of oxygen occurs from the capillary blood into the water.



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 20** A female child is affected by a condition caused by a recessive allele of a gene located on the X chromosome.

Neither of her biological parents has the condition.

Genetic testing shows that every somatic cell in the child contains only one copy of this gene, which is the mutant allele.

Which of the following statements could explain this?

- 1 Non-disjunction of sex chromosomes occurred during meiosis in her father.
- 2 A deletion of the gene occurred on the X chromosome during meiosis in her father.
- 3 The mutant allele was inherited from an affected paternal grandfather.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 21** A population of insects possesses a gene with two alleles, A^* and a^* . The table below shows the probability of survival to reproductive age for each genotype in two different environments. | Genotype | Environment 1 | Environment 2 | |---|---|---| | A^*A^* | 0.60 | 0.95 | | A^*a^* | 0.95 | 0.95 | | a^*a^* | 0.00 | 0.00 | The population has lived in Environment 1 for many generations and the frequency of allele a^* is stable. The environment then changes permanently to resemble Environment 2.

Which of the following describes the expected change in the frequency of allele a^* over the subsequent generations?

- A** It will remain constant, because the survival rate of the heterozygote Aa remains high.
- B** It will increase, because the selection pressure against the homozygous dominant genotype AA is removed.
- C** It will decrease, because the selective advantage of the heterozygote is lost while the lethal cost of aa remains.
- D** It will decrease to zero in a single generation, because individuals with the aa genotype cannot reproduce.

- 22** A continuous segment of double-stranded DNA is 300 base pairs long. In this DNA segment, 36% of the nitrogenous bases are thymine (T).

The entire DNA segment is transcribed into an mRNA molecule of 300 nucleotides. Translation of this mRNA begins at the first codon and ends at a stop codon located at the final triplet of the sequence.

Which row in the table correctly gives the total number of hydrogen bonds between the bases in this DNA segment and the number of peptide bonds formed in the resulting polypeptide?

	total number of hydrogen bonds in DNA segment	number of peptide bonds in polypeptide
A	342	98
B	342	99
C	684	98
D	684	99
E	684	100
F	816	98
G	816	99
H	816	100

- 23 The table shows some mRNA codons and the amino acids or translation signals they encode:

mRNA codon	Amino acid or function
AUG	Met (Start)
CAA	Gln
CAG	Gln
GAA	Glu
GAG	Glu
GGA	Gly
GGC	Gly
UUA	Leu
UUG	Leu
UCA	Ser
UCG	Ser
UGG	Trp
UAA	Stop
UAG	Stop
UGA	Stop

An unmutated mRNA molecule has the sequence:

5'–AUG UUA GAA GGA UCG UGG GAG CAA UAA–3'

Single nucleotide substitution mutations took place at both the 8th and 17th nucleotides in this sequence.

Using **only** the information provided, which of the following statements **could** be correct for the polypeptide produced by translating this mutated sequence from left to right?

- 1 The resulting polypeptide could be only five amino acids long.
- 2 The resulting polypeptide could contain only four different types of amino acids.
- 3 The resulting polypeptide could contain seven different types of amino acids.

- A** none of them
B 1 only
C 2 only
D 3 only
E 1 and 2 only
F 1 and 3 only

- G** 2 and 3 only
H 1, 2 and 3

24 Researchers are investigating a homeostatic defect in a mutant strain of mice. They compare the blood glucose concentration of mutant and wild-type mice under two experimental conditions:

1. **Glucose Tolerance Test:** Mice are injected with a glucose solution.
2. **Fasting Test:** Mice are withheld from food for 12 hours.

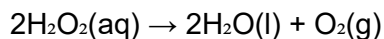
The results are summarised below:

Condition	Wild-type response	Mutant response
Glucose Tolerance Test	Blood glucose rises, then returns to baseline within 90 minutes	Blood glucose rises, then returns to baseline within 90 minutes
Fasting Test	Blood glucose remains stable within the normal range	Blood glucose decreases rapidly to hypoglycaemic levels

Which of the following defects best explains these results?

- A** The mutant mice are unable to synthesise insulin.
B The mutant mice have insulin receptors that are permanently activated.
C The mutant mice constitutively secrete high levels of insulin.
D The mutant mice are unable to secrete glucagon.
E The mutant mice have glucagon receptors that are permanently activated.

- 25** A student investigated the rate of an enzyme-catalysed reaction using catalase to break down hydrogen peroxide into water and oxygen gas:



Under standard control conditions, 10 cm³ of 2.0% hydrogen peroxide solution was mixed with 1.0 cm³ of catalase solution at 25 °C and pH 7.0. The volume of oxygen gas produced over time was recorded, giving curve **X** on the graph.

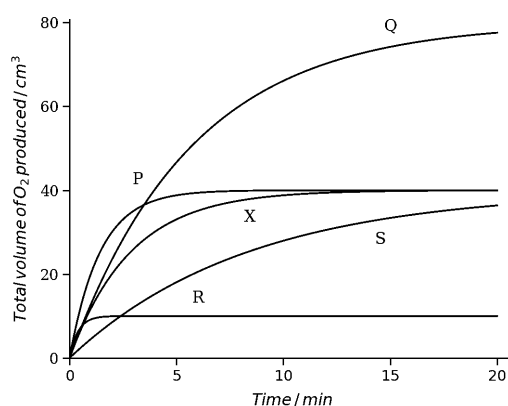
The student then carried out four separate variations of the experiment:

- **Variation 1:** 10 cm³ of 2.0% hydrogen peroxide solution mixed with 2.0 cm³ of catalase solution at 25 °C and pH 7.0.
- **Variation 2:** 20 cm³ of 2.0% hydrogen peroxide solution mixed with 1.0 cm³ of catalase solution at 25 °C and pH 7.0.
- **Variation 3:** 10 cm³ of 2.0% hydrogen peroxide solution mixed with 1.0 cm³ of catalase solution at 75 °C and pH 7.0.
- **Variation 4:** 10 cm³ of 2.0% hydrogen peroxide solution mixed with 1.0 cm³ of catalase solution at 10 °C and pH 7.0.

Assume the catalase has an optimum temperature of 37 °C and denatures rapidly above 60 °C.

Which row in the table correctly identifies the curve corresponding to each variation?

	Variation 1	Variation 2	Variation 3	Variation 4
A	P	Q	R	S
B	P	Q	S	R
C	P	S	R	Q
D	Q	P	R	S
E	Q	S	P	R
F	S	P	R	Q
G	S	Q	P	R
H	S	Q	R	P



- 26** Cylinders of potato tissue were immersed in sucrose solutions for 24 hours. The results are shown below.

Water potential of solution, Ψ / MPa	Mean change in mass / %
−0.2	+12
−0.5	+6
−0.8	0
−1.1	−6

At no net mass change, the tissue water potential equals the solution water potential. For this question, cell water potential is given by $\Psi = \Psi_s + \Psi_p$, where Ψ_s is solute potential and Ψ_p is pressure potential.

At equilibrium, $\Psi_p = 0.25$ MPa. What is Ψ_s ?

- A** −0.55 MPa
B −0.80 MPa
C −1.05 MPa
D −1.35 MPa
- 27** A unicellular organism was analysed and found to possess the following features:
- * circular double-stranded DNA not enclosed by a nuclear envelope
 - * 70S ribosomes in the cytoplasm
 - * no membrane-bound organelles
 - * a surrounding cell wall
- Which of the following statements is/are correct?
- 1 It cannot carry out aerobic respiration.
 - 2 It possesses genes that code for ribosomal RNA.
 - 3 It reproduces asexually by binary fission rather than mitosis.
- A** none of them
B 1 only
C 2 only
D 3 only
E 1 and 2 only
F 1 and 3 only
G 2 and 3 only
H 1, 2 and 3